

RVHS JC2 H2 Physics Prelim Examination 3 Mark Scheme

1	(a)	(i)	time = $0.43 / 1.1$ = 0.39(1) s	A1
		(ii)	$s = ut + \frac{1}{2}at^2$ $= \frac{1}{2}(9.81)(0.39)^2$ = 0.75(0) m	C1 A1
		(iii)	vertical velocity: $v = u + at$ $= (9.81)(0.39)$ $= 3.8259 \text{ m s}^{-1}$	M1
			$\theta = \tan^{-1} \frac{v_y}{v_x} = \tan^{-1} \frac{3.8259}{1.1}$ = 74(.0)°	A1
		(iv)	1. Horizontal line at a non-zero value of a.	B1
			2. Curved line from origin with increasing gradient	B1
	(b)		acceleration of free fall is <u>unchanged</u> / not dependent on mass <u>and</u> so no effect (on time taken)	A1

2	(a)		T_1 and T_2 : down	B1
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